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PROJECT AND TEAM INFORMATION

Project Title

Sanchar: A Bluetooth-Based Chat Application

Team Name: *Team #:* *Team Lead:*	Tech Titans DSCPP-III-2026-T344 Divya Jain
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PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

Sanchar is an offline, peer-to-peer text-chat application for two paired Windows laptops. It uses Bluetooth RFCOMM to exchange messages without internet, Wi-Fi, mobile data or a central server. The application supports device scanning, pairing, hosting and connecting, printable ASCII messages of 1 to 1023 characters, and persistent conversation history stored separately for each device. C implements circular message queues, linked-list history, insertion sort and linear search. C++ classes organize the application and communication modules. A practice mode provides simulated replies for demonstration without a second laptop. Phase II includes the implemented modules and initial two-laptop validation; further testing and performance measurements remain pending.

Updated Project Approach and Architecture

The system follows a modular design. The application module in main.cpp handles menu choices, keyboard input and the chat display. The Bluetooth module in bluetooth.cpp/.hpp performs scanning, pairing, hosting and connecting through Windows Winsock2 (ws2bth) and the Windows Bluetooth API. Peers locate the Sanchar service using a UUID.

The chat module in chat.hpp contains Message, MessagePump and ChatHistory. MessagePump uses outgoing and incoming circular queues, newline message framing and a send offset to handle partial transfers. Nonblocking sockets and keyboard polling keep input responsive. Complete messages are displayed and saved in per-device plain-text files under history/.

The C DSA module supplies queues, a singly linked list with a tail pointer, device sorting and searching. BluetoothConnection and DemoConnection implement the abstract ConnectionHandler interface, allowing the same chat logic to use either real Bluetooth or simulated practice replies. The finalized scope is one-to-one chat on Windows; no DBMS is used.

Tasks Completed

Task Completed	Team Member
Message validation, queued send/receive processing and per-device saved history in chat.hpp.	Divya Jain
Device scanning with paired status and Windows Bluetooth pairing in bluetooth.cpp; device record in bluetooth.hpp.	Abhay Pratap Singh
Main menu, chat screen, command handling and simulated practice mode in main.cpp and demo.hpp.	Prince Awasthi
Circular queues, linked-list history, insertion sort and linear search in dsa.c / dsa.h.	Dhirendra Kishor Shah
Host/connect functionality and authenticated RFCOMM sockets implemented.	Individual ownership not specified

Challenges/Roadblocks

Bluetooth stream data does not preserve message boundaries. Newline framing identifies complete messages when data arrives in partial or combined chunks. Nonblocking sends may transmit only part of a message, so a send offset records the next byte to send. Fixed-size queues bound memory usage and provide O(1) push/pop operations. Service discovery uses a UUID instead of a hard-coded Bluetooth channel.

The Windows-specific Bluetooth implementation limits portability. Remaining validation includes fuller two-laptop testing, history reload verification, and response-time and reliability measurements.

Tasks Pending

Task Pending	Proposed Responsibility
Complete two-laptop connection testing and record response time and reliability.	Abhay Pratap Singh; Dhirendra Kishor Shah
Verify history reload and message order across repeated sessions.	Divya Jain
Show peer display names instead of Bluetooth addresses.	Abhay Pratap Singh; Prince Awasthi
Finalize deployment, demonstration and user documentation.	All team members; coordinated by Divya Jain

Project Outcome/Deliverables

The project deliverables comprise a console-based offline chat application for two paired Windows laptops; Bluetooth scanning, pairing, hosting and connecting; validated text messaging; timestamped, per-device local history; and a simulated practice mode. The C/C++ codebase demonstrates queues, linked lists, searching, sorting, classes, inheritance and an abstract connection interface.

Expected final outcomes include measured two-laptop test results, final deployment, a working demonstration and user documentation. The application is intended for nearby communication where internet access is unavailable. Its current scope is one-to-one chat on Windows.

Progress Overview

Code has been written for the Bluetooth, chat, DSA and demo/UI modules. The prototype includes scanning, pairing, host/connect operations, timestamped messages, /history and /quit commands, local history and practice mode. Initial results show successful scanning, pairing, connection and two messages received in order in one direction.

History validation is partial: the saved file contains both messages, but reload was not demonstrated. Further two-laptop testing, response-time and reliability measurements, peer display names, final deployment and documentation remain. The repository activity shown is eight commits: five by Abhay and three by Dhirendra.

Completion assessment: Core implementation is complete at the code level; the project is in the integration and validation stage. Final readiness depends on repeatable test results and documentation.

Proposed next-stage sequence: Verify bidirectional messaging and history reload, measure response time and reliability, improve peer identification, then prepare the final demonstration. Schedule variance will be assessed against agreed Phase-III dates.

Codebase Information

Repository: https://github.com/abhaypratapsingh8273-art/sanchar
Visibility: Public. Default branch: main. Branches: main only.
Recorded development activity: 8 commits in the Phase-II repository snapshot.
Key files: bluetooth.cpp, bluetooth.hpp, chat.hpp, connection.hpp, demo.hpp, dsa.c, dsa.h, main.cpp and README.md.
Development coverage: Bluetooth connectivity, message processing, C data structures, practice mode and the application interface.
Proposed workflow: Record future changes with descriptive commits and review them before merging into main.

Testing and Validation Status

Test Type	Status	Notes
Scan (Bluetooth)	Pass	6 devices listed; paired status shown.
Pair (Bluetooth)	Pass	"Pairing completed" for device 3.
Host / Connect	Pass	Both laptops entered a chat session.
Send/receive	Pass	2 messages received in order in one direction.
History	Partial	Saved file contains both messages; reload not shown.

Planned reliability measurement: Send numbered messages in both directions and record received, missing and duplicate messages across repeated trials.
Planned response-time measurement: Record elapsed send-to-receive time using a consistent timing method. Measured values remain pending.

Deliverables Progress

Completed (code): Bluetooth, chat, DSA and demo/UI modules, including the console interface and practice mode.

In progress: Integrated two-laptop validation and local-history verification. Initial connection and messaging tests passed; history reload remains unverified.

Pending: Measured response time and reliability results, peer display names, final deployment, demonstration and user documentation. Proposed completion checks are repeatable bidirectional messaging, successful history reload and a documented test summary.